A Low-Complexity CSI-Based Wifi Sensing Scheme for LoS-Dominant Scenarios

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***Abstract*—Integrating sensing into wifi platforms, referred to as wifi sensing, provides an efficient and device-free means for indoor monitoring/localization with low cost. Clock asynchrony is one of the most challenging issues in wifi sensing. A mainstream solution to date employs cross-antenna processing to suppress the clock offsets that are common to all antennas. Such methods, however, may suffer from issues such as mirrored targets and noise enhancement etc. This paper develops a novel wifi sensing scheme. It embodies accurate and low-complexity methods for estimating the timing and frequency offsets as well as the angle-of-arrival (AoA), all from the LoS path. It also involves a coherent Doppler processing method which effectively suppresses static paths and accurately estimates the Doppler by enjoying the coherent processing gain. Corroborated by experimental results using open Widar2.0 data, the proposed design is able to precisely recover the velocity traces in various indoor scenarios.**

***Index Terms*—Integrated sensing and communications (ISAC), wifi sensing, OFDM, channel state information, Doppler, LoS**

I. Introduction

Wifi indoor sensing has become extremely popular recently [1]–[5]. It can sense vital health signs and body gestures by simply reusing wifi frequency spectrum and needing no on-body devices. Also, sensing can be integrated into wifi platforms, with additional cost mainly in firmware upgrading. Furthermore, future wifi transceiver can have multiple antennas and cover multiple frequency bands, which potentially enables high-quality sensing. However, all these benefits do not come without a price.

A wifi transceiver generally works in a half-duplexing mode; that is, when its transmitter is on (off), its receiver stays off (on). This means at least two wifi transceivers are generally required for wifi sensing. Since low-cost clocks are used in wifi devices, the clock asynchrony issue is predominant in wifi sensing. Though a wifi receiver (Rx) would generally perform clock synchronization using the training signals from a wifi transmitter (Tx), the level of synchronization, which is sufficient for data communications, can still cause unacceptable sensing errors. For instance, the frequency offset residual in a wifi Rx can be in the order of 10 Hertz (Hz). This would lead to an Doppler estimation error in the order of $0.515 (= 10 \times \frac{3\times 10^8}{5.825\times 10^9})$ meter per second (m/s), considering the 5.825 GHz wifi channel.

To solve the synchronization issue in wifi sensing, numerous solutions have been developed in the literature [6]. For a single wifi receiver with multiple antennas, a mainstream technique is the cross-antenna processing, as the receiver antennas, which share the same clock, have common clock offsets with wifi Tx. Specifically, cross-antenna cross-correlation (CACC) and cross-antenna signal (or CSI) ratio (CASR) have been developed. CACC removes timing and frequency offsets by taking the cross-correlation between the signals, or channel state information (CSI), from different wifi-Rx antennas [7], [8]. A main challenge of this method is that the image components generated through the cross-correlation is not easy to remove. A power adjustment strategy is devised in [7] to increase the power of the term containing the correct Doppler frequency information for better identification in the Doppler spectrum. Other alternative schemes, such as the mirrored MUSIC algorithm [8], have also been proposed.

CASR removes clock offsets by taking the ratios of CSIs on different antennas. It does not generate the image components as do CACC and can be simpler in implementation [9]. However, CASR has mainly been demonstrated in detecting respiration patterns [9], [10] and simple straight-line motion [11], leaving complex scenarios with irregular and fast movement unexplored. Moreover, when CSI is close to zero, as is possible in rich scattering cases, taking the ratio can incur severe noise enhancement, degrading the overall sensing performance.

The cross-antenna processing can suppress common clock offsets over all antennas, which, however, also makes the echo model complicated. This further limits the application of cross-antenna-based methods in sensing multiple targets [12]. In this work, we develop a novel wifi sensing scheme without resorting to the cross-antenna techniques. In particular, assuming the dominance of the line-of-sight (LoS) path, we design new methods to accurately estimate the timing and frequency offsets using the LoS path. Compensating the offsets on all paths enables us to jointly utilize multiple consecutive symbols for moving target detection and estimation. Our main contributions are summarized as follows.

1) We design a low-complexity and accurate estimation method for the composite time delay of the LoS path which is the sum of the LoS propagation delay and timing offset. The method does not involve time-consuming search over fine delay grids, leading to the off-grid accurate delay estimation.
2) We also develop accurate estimators for the frequency offsets and the angle-of-arrival (AoA) of the LoS path. The proposed methods are able to obtain accurate AoA estimation even with the presence of moving target. Thus, our design removes the need of target-free training for the AoA estimation as required by the state of the art [13].
3) Enabled by the proposed estimation methods for the

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LoS parameters, we further design a coherent Doppler processing scheme which first suppresses all static paths and then jointly utilizes multiple consecutive symbols for accurate Doppler frequency estimation.

We employ the open data Widar2.0 [13] for validating the proposed wifi sensing scheme. For various scenarios, the proposed wifi sensing scheme is able to accurately estimate the LoS parameters, effectively remove timing and frequency offsets, and precisely recover the velocity traces highly matching the results deduced from the ground truths.

In the remaining of the work, Section II establishes the signal model of wifi sensing. Section III presents the proposed wifi sensing scheme. Section IV validates the proposed scheme through experimental results, followed by conclusions in Section V.

II. Signal Model

In this section, we establish the signal model of indoor wifi sensing, illustrating the impact of human motion on wifi CSIs. For simplicity, we consider a single-input and multiple-output (SIMO) system. In multiple-input and multiple-output (MIMO) wifi, different transmitting antennas would use orthogonal resources for CSI acquisition. Thus, the proposed methods based on SIMO (with a single tansmitting antenna) can be readily extended to MIMO cases.

Consider an $N_{\rm a}$-antenna uniform linear array (ULA) at a wifi receiver. It receives P multiple paths each with the angle of arrival (AoA) θ_p. Let f_0 denote the sub-carrier frequency interval, τ_p denote the propagation delay of path p, and μ_p denote the Doppler frequency of the path. The CSI vector at the n-th sub-carrier and m-th OFDM symbol can be given by

$$\mathbf{h}_{n,m} = \sum_{p=0}^{P-1} \alpha_l e^{\mathrm{j}2\pi n f_0(\tau_l+\epsilon_m^\tau)} e^{\mathrm{j}2\pi(\mu_l+\epsilon_m^f)mT} \mathbf{a}_{N_{\rm a}}(\theta_l), \quad (1)$$

where α_l denotes the path gain, ϵ_m^τ is a time-varying timing offset, and ϵ_m^f is a time-varying frequency offset. Here, we assume that the intra-symbol impact of Doppler frequency and carrier frequency offset can be neglected. Moreover, $\mathbf{a}_{N_{\rm a}}(\theta_l)$ denotes the receiving array steering vector. Its i-th entry can be given by

$$[\mathbf{a}_{N_{\rm a}}(\theta_l)]_i = e^{\mathrm{j}2\pi i d \sin(\theta_l)/\lambda},\ i = 0, 1, \cdots, N_{\rm a}-1, \quad (2)$$

where d denote the antenna spacing and λ is the wavelength. Note that $[\cdot]_i$ takes the i-th entry of the enclosed vector.

In this work, we study the estimation of the Doppler frequency of a single human moving indoors in the presence of the line-of-sight (LoS) path between wifi transceiver.

III. Proposed Doppler Estimation

For illustration convenience, we let the path $p = 0$ be the LoS path, and the path $p = 1$ be the single moving target. Thus, we can rewrite $\mathbf{h}_{n,m}$ into (3).

The following two terms are introduced for convenience,

$$\begin{array}{ll} \tilde{\tau}_m^0 = \tau_0 + \epsilon_m^\tau & \text{(composite LoS time delay)} \\ \tilde{\alpha}_m^0 = \alpha_0 e^{\mathrm{j}2\pi\epsilon_m^f mT} & \text{(composite LoS gain)} \end{array}, \quad (4)$$

$$\begin{array}{ll} \mathbf{h}_{n,m} = \alpha_0 e^{\mathrm{j}2\pi n f_0(\tau_0+\epsilon_m^\tau)} e^{\mathrm{j}2\pi\epsilon_m^f mT} \mathbf{a}_{N_{\rm a}}(\theta_0) & \text{(LoS path)} \\ +\alpha_1 e^{\mathrm{j}2\pi n f_0(\tau_1+\epsilon_m^\tau)} e^{\mathrm{j}2\pi(\mu_1+\epsilon_m^f)mT} \mathbf{a}_{N_{\rm a}}(\theta_1) & \text{(moving target)} \\ +\sum_{p=2}^{P-1} \alpha_l e^{\mathrm{j}2\pi n f_0(\tau_l+\epsilon_m^\tau)} e^{\mathrm{j}2\pi\epsilon_m^f mT} \mathbf{a}_{N_{\rm a}}(\theta_l). & \text{(static paths)} \end{array} \quad (3)$$

where $\tilde{\tau}_m^0$ is affected by the timing offset, and $\tilde{\alpha}_m^0$ incorporates the impact of frequency offset. In general, the LoS path is stronger than other paths. Utilizing this feature, we propose to estimate **first** the composite timing and frequency offsets using the LoS path. **Then**, we compensate the two offsets for all paths. **Third**, we suppress static paths and estimate the Doppler frequency of the moving target in the Doppler domain. Novel methods are developed next to perform the three major steps.

A. Estimate LoS Parameters

We start with developing a low-complexity method for estimating the composite LoS time delay $\tilde{\tau}_m^0$. Then, we derive the composite frequency offset $\alpha_0 e^{\mathrm{j}2\pi\epsilon_m^f mT}$ from the reference antenna. The overall methods are summarized in Algorithm 1.

Note that ϵ_m^τ in $\tilde{\tau}_m^0$ can change over m; see (4). Thus, we estimate $\tilde{\tau}_m^0$ for each symbol m independently. Based on (3), we rewrite the i-th entry of $\mathbf{h}_{n,m}$ into the following convenient form,

$$h_{m,i}[n] = \beta_{m,i} e^{\mathrm{j}n\omega_m^0} + \xi_{m,i},\ \text{s.t. } \omega_m^0 = 2\pi f_0 \tilde{\tau}_m^0 \quad (5)$$

where $\beta_{m,i}$ simply absorbs all the coefficients irrelevant of $\tilde{\tau}_m^0$, and $\xi_{m,i}$ collects all the non-LoS paths and noise. Note that $h_{m,i}[n]$ can be seen as a discrete noisy single-tone signal with the angular frequency of $\omega_m^0 = 2\pi f_0 \tilde{\tau}_m^0$. Estimating ω_m^0 from $h_{m,i}[n]$ can be done through numerous methods, such as the maximum likelihood, super-resolution and even compressive sensing methods. However, considering that the number of samples can be limited and to obtain an off-grid estimation with low complexity, we modify our recently developed frequency estimator [14] for the task here. We remark that non-trivial efforts are made for this modification, as detailed next.

Since $h_{m,i}[n]$ is a single-tone signal, its discrete Fourier transform (DFT) would show a single mainlobe with a peak. The frequency bin where the peak locates gives an approximate frequency estimate with an error up to half the discrete frequency bin, i.e., $\frac{\pi}{N}$, where N is the DFT dimension, also the sub-carrier number. Moreover, $h_{m,i}[n]$ also varies with i, the index of antennas at a wifi receiver. In narrowband cases, the difference in propagation time over receiver antennas is negligible compared with the LoS time delay. Also, the timing offset is the same on all antennas (each connected to the same local oscillator). Thus, we perform an incoherent accumulation of the magnitude of the DFT results over the antennas to improve the detecting probability of the peak index. This is summarized in Steps 1)-3) of Algorithm 1. The obtained peak index is denoted by k (starting from zero). Since $\mathsf{k} \pm 1$ will be used next, we also assume that $\mathsf{k} \geq 1$ and $\mathsf{k} \leq N-2$; otherwise, we will need to use a modulo function to revise the result, e.g., turning -1 into $N-1$.

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$$
\begin{aligned}
a_1 &= -\frac{15\pi\sqrt{2}N^2\sigma_2\left(-2N^4\sigma_2^4+14N^3\sigma_1\sigma_2^3+24N^2\sigma_1^4-24N^2\sigma_1^2+15N\sigma_1^3\sigma_2-N\sigma_1\sigma_2^3-4\sigma_1^4+\sigma_1^2\right)}{4} \quad \text{where } \sigma_1{=}\cos\left(\tfrac{\pi}{4N}\right);\ \sigma_2{=}\sin\left(\tfrac{\pi}{4N}\right)\\
a_3 &= -\frac{7\sqrt{2}\pi^3N^6\sigma_2^5}{2}-\frac{3\sqrt{2}\pi^3N^5\sigma_1\sigma_2^4}{2}+30\sqrt{2}\pi^3N^4\sigma_1^2\sigma_2^3-60\sqrt{2}\pi^3N^3\sigma_1^3\sigma_2^2+10\sqrt{2}\pi^3N^3\sigma_1\sigma_2^4+45\sqrt{2}\pi^3N^2\sigma_1^4\sigma_2-15\sqrt{2}\pi^3N^2\sigma_1^2\sigma_2^3\\
&\quad -\frac{45\sqrt{2}\pi^3N\sigma_1^5}{4}+\frac{45\sqrt{2}\pi^3N\sigma_1^3\sigma_2^2}{4}-\frac{3\sqrt{2}\pi^3N\sigma_1\sigma_2^4}{2}-\frac{15\sqrt{2}\pi^3\sigma_1^4\sigma_2}{4}+\frac{\sqrt{2}\pi^3\sigma_1^2\sigma_2^3}{4} \qquad (6)\\
b_0 &= -\frac{15\sqrt{2}N^3\sigma_2^2\left(2N^3\sin\left(\frac{\pi}{4N}\right)^3-12N^2\sigma_1\sigma_2^2+12N\sigma_1^2\sin\left(\frac{\pi}{4N}\right)-3\sigma_1^3+\sigma_1\sigma_2^2\right)}{8}\\
b_2 &= -\frac{3\sqrt{2}N\pi^2\left(2N^5\sigma_2^5-80N^4\sigma_1\sigma_2^4-280N^3\sigma_2^5+280N^3\sigma_2^3+300N^2\sigma_1^5-300N^2\sigma_1^3+20N^2\sigma_1\sigma_2^4+120N\sigma_1^4\sigma_2+40N\sigma_2^5-40N\sigma_2^3-30\sigma_1^5+15\sigma_1^3-2\sigma_1\sigma_2^4\right)}{8}
\end{aligned}
$$

The true angular frequency can be smaller or larger than $2\pi\mathsf{k}/N$. Through a sign test [14], we can further tell whether it is smaller or larger. This is performed in Steps 4)-5). With α obtained in Step 5), we know that the true frequency is in the range of $\left[\frac{2\pi\mathsf{k}}{N}, \frac{2\pi(\mathsf{k}+\alpha)}{N}\right]$ if $\alpha = 1$; otherwise, it is in $\left[\frac{2\pi(\mathsf{k}+\alpha)}{N}, \frac{2\pi\mathsf{k}}{N}\right]$. The residual is then estimated to refine the angular frequency estimate. To do this, we construct the following ratio,

$$\rho = (x_+^2 - x_-^2)\Big/(x_+^2 + x_-^2), \qquad (7)$$
$$\text{s.t. } x_+ = \tilde{H}_m[\max\{\mathsf{k}, \mathsf{k}+\alpha\}],\ x_- = \tilde{H}_m[\min\{\mathsf{k}, \mathsf{k}+\alpha\}],$$

where $\tilde{H}_m[\mathsf{k}]$ is obtained in Step 2) of Algorithm 1, and it is already a magnitude. From ρ, we can solve the residual. First, we calculate the following coefficients,

$$k_2 = -\rho b_2/a_3, k_1 = a_1/a_3, k_0 = -\rho b_0/a_3, \qquad (8)$$

where a_1, a_3, b_0 and b_2 are intermediate variables as given in (6). They only need to be calculated once given the value of N. We note that the analytical forms of the coefficients in (6) are newly derived in this work.

Using the k-coefficients obtained above, we further calculate the following,

$$\begin{aligned} z_1 &= -k_2/3 + 2B,\ z_2 = -k_2/3 - B + E,\\ z_3 &= -k_2/3 - B - E, \end{aligned} \qquad (9)$$

with the intermediate variables given by

$$B = (S+T)/2,\ E = \sqrt{3}(S-T)\mathrm{j}/2;$$
$$S = \sqrt[3]{R+\sqrt{D}},\ T = \sqrt[3]{R-\sqrt{D}},\ D = Q^3 + R^2;$$
$$R = \frac{(9k_1k_2 - 27k_0 - 2k_2^3)}{54},\ Q = \frac{(3k_1 - k_2^2)}{9}.$$

Though (6)-(9) seem complicated, they only involve elementary arithmetic operations. Thus, they can be computed even in low-performance computing platforms. Among z_1, z_2 and z_3, the one with the smallest magnitude can be used for refining the frequency estimate, as given by

$$\widehat{\omega_m^0} = \frac{2\pi(\mathsf{k}+\delta)}{N},\ \text{s.t. } \delta = \delta + \Re\{z_{i^*}\},\ i^* = \arg\min_i |z_i|, \qquad (10)$$

where δ is zero initially. One may refer to [14] for the rationale behind (9)-(10).

An iterative update can be further performed to refine $\widehat{\omega_m^0}$. In particular, we first calculate the following interpolated DFT coefficients,

$$H_\pm = \sum_{n=0}^{N-1} h_{m,i}[n] e^{-\mathrm{j}\frac{2\pi n(\mathsf{k}+\delta\pm 0.25)}{N}}, \qquad (11)$$

Algorithm 1 Estimation of LoS parameters

Initialize: Set $\delta = 0$, $i_{\text{iter}} = 0$, I_{iter},
Estimate $\widehat{\tau_m^0}$, the composite LoS time delay.
1) Calculate the DFT of $h_{m,i}[n]$, leading to $H_{m,i}[k]$;
2) For each k, sum $|H_{m,i}[k]|$ over i, leading to $\tilde{H}_m[k]$;
3) Get k such that $\tilde{H}_m[\mathsf{k}]$ is the maximum;
4) For each i, calculate $\alpha_i = \text{sign}\{[H_{m,i}(\mathsf{k}-1) - H_{m,i}(\mathsf{k}+1)]H_{m,i}^\dagger(\mathsf{k})\}$, where $()^\dagger$ takes the conjugate;
5) Set $\alpha = \text{sign}\left\{\sum_{i=0}^{N_\text{a}} \alpha_i\right\}$;
6) Calculate ρ as in (7);
7) Calculate the k-coefficients per (8);
8) Calculate the three z's in (9)
9) Update $\widehat{\omega_m^0}$ and δ as given in (10);
10) If $i_{\text{iter}} < N_{\text{iter}} - 1$, proceed; otherwise, go to Step 12).
11) Calculate $H_\pm$ per (11) and ρ as in (12); go back to Step 7).
Estimate the composite frequency offset per (14).
Estimate LoS-AoA: reset $\delta = 0$ and $i_{\text{iter}} = 0$
12) Compensate timing offset on all antennas, yielding $s_m[i]$ in (15);
13) With $h_{m,i}[n]$ replaced by $s_m[i]$, rerun Steps 1)-11), resulting in the estimate of $2\pi d\sin(\theta_0)/\lambda$, as denoted by $\hat{u}_m$ for symbol m.

where k is obtained in Step 5) of Algorithm 1 and the updated δ in (10) is used. Similar to (7), we update ρ as

$$\rho = (|H_+|^2 - |H_-|^2)\Big/(|H_+|^2 + H_-|^2). \qquad (12)$$

Then, we redo the calculations in (8)-(10) to update $\widehat{\omega_m^0}$. After convergence; see Step 10) for possible conditions, we obtain the final frequency estimate. Referring to (5), we can estimate the composite time delay of the LoS path as

$$\widehat{\tau_m^0} = \widehat{\omega_m^0}/(2\pi f_0). \qquad (13)$$

Moreover, combining (5) and (2), we can compensate the composite time delay on the reference antenna (with $i = 0$) and obtain the estimate of the composite frequency offset, i.e.,

$$\widehat{\alpha_0 e^{\mathrm{j}2\pi\epsilon_m^f mT}} = \frac{1}{N}\sum_{n=0}^{N-1} h_{m,0}[n] e^{-\mathrm{j}n\widehat{\omega_m^0}}. \qquad (14)$$

We note that the accuracy of the estimation obtained in (14) would be dependent on the dominance of the LoS path.

Similar to (14), we can compensate the LoS time delay at each symbol for other antennas. The resulted signal can be written as

$$\begin{aligned} s_m[i] &= \frac{1}{N}\sum_{n=0}^{N-1} h_{m,i}[n] e^{-\mathrm{j}n\widehat{\omega_m^0}}\\ &\approx \widehat{\alpha_0 e^{\mathrm{j}2\pi\epsilon_m^f mT}} e^{\mathrm{j}2\pi i d\sin(\theta_0)/\lambda},\ i = 0, 1, \cdots, N_\text{a} - 1, \end{aligned} \qquad (15)$$

where the approximation is due to the removal of all other paths. Again, the validity of the removal is built on the

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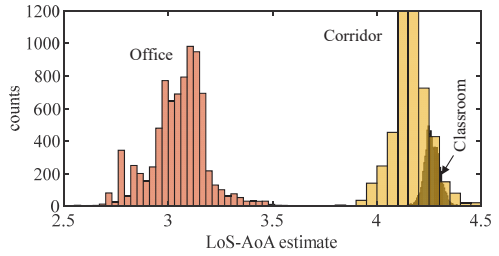


Fig. 1: Histograms of $\hat{u}_m$'s obtained in Step 13) of Algorithm 1, where the unit is radian.

stronger LoS path. Note that the coefficient of the i-dependent exponential term in (15) is independent of i. Thus, $s_m[i]$ can be seen as a single-tone signal; similar to $h_{m,i}[n]$ obtained in (5). Conveniently, we can treat $2\pi d\sin(\theta_0)/\lambda$ as an angular frequency and apply the method developed for the LoS time delay estimation. This is performed in Steps 12)-13) of Algorithm 1. Each symbol can generate an estimate, as denoted by $\hat{u}_m$ in the algorithm.

Ideally, $\hat{u}_m$ $(\forall m)$ should be the same under stationary wifi transceiver. However, affected by noise and the estimation errors in the composite LoS time delays, $\hat{u}_m$ present some randomness over m. This is demonstrated in Fig. 1. The figure is obtained by performing the proposed methods using the open experimental data Widar2.0 [13]. Details about the data will be given in Section IV. Here, we only want to draw some insights related to the methods developed in this section.

In different scenarios, several thousands of OFDM symbols are used in performing Algorithm 1. Despite the spread of AoA estimates, we can see that they are centered at some values with Gaussian-like distributions. This validates the feasibility of the proposed LoS estimation scheme under practical scenarios, even though we have assumed strong LoS-dominance during the algorithm development. We highlight the fact that the AoA estimation is performed by compensating the composite LoS delays overall thousands of symbols which are estimated using the proposed method. Therefore, the effectiveness of the proposed delay estimation method is also manifested by Fig. 1; otherwise, we would not obtain such concentrated AoA estimates.

B. Estimate the Doppler Frequency of the Moving Target

Using the estimates obtained in (13) and (14), we can suppress the timing and frequency offsets in $\mathbf{h}_{n,m}$ through

$$\tilde{\mathbf{h}}_{n,m} = \mathbf{h}_{n,m}\left(\alpha_0 e^{\mathrm{j}2\pi\widehat{\epsilon f_m}mT}\right)^{\dagger} e^{-\mathrm{j}n\widehat{\omega_m^0}} \tag{16}$$

$$\approx \begin{array}{ll} |\alpha_0|^2\mathbf{a}_{N_\mathrm{a}}(\theta_0) & \text{(LoS path)} \\ +\alpha_1\alpha_0^{\dagger}e^{\mathrm{j}2\pi n f_0(\tau_1-\tau_0)}e^{\mathrm{j}2\pi\mu_1 mT}\mathbf{a}_{N_\mathrm{a}}(\theta_1) & \text{(moving target)} \\ +\sum_{p=2}^{P-1}\alpha_l\alpha_0^{\dagger}e^{\mathrm{j}2\pi n f_0(\tau_l-\tau_0)}\mathbf{a}_{N_\mathrm{a}}(\theta_l) & \text{(static paths)} \end{array},$$

where the approximation holds when the estimation of LoS parameters are reasonably accurate. An important observation from (16) is that the moving target is now the only one with a non-zero Doppler frequency. This suggests that we may separate the moving target with the static ones, enabling the estimation of its Doppler frequency. This is similar to the moving target detection in traditional radar processing; however, this only becomes valid here owing to our proposed methods in estimating the LoS parameters.

The approach adopted here is to filter out the static paths first and then estimate the Doppler frequency of the remaining moving target. The imperfect estimation of LoS parameters may lead to the spread of static targets around zero frequency in the Doppler spectrum. To this end, we can employ a notch filter with a certain null width to suppress them more effectively. To determine the null width, we quantify the impact of the notch filter on Doppler measurements.

From (16), we see that the sampling interval in the (slow) time domain is T. Thus, the frequency range of the Doppler domain is $\left[-\frac{1}{2T},\frac{1}{2T}\right]$. Moreover, depending on the up limit of m, as denoted by M, we can get a Doppler resolution of $\frac{1}{MT}$. Given $\mu=\frac{v}{\lambda}$, where μ denotes the Doppler frequency in this work, the measurable velocity resolution is

$$\Delta_v = \lambda/(MT). \tag{17}$$

Therefore, we come to the following conclusion: *If the notch filter has a null width of N_0 times Doppler frequency bins, the velocity between $\left[-\frac{N_0\lambda}{2MT},\frac{N_0\lambda}{2MT}\right]$ (m/s) will not be measurable.*

Let us check a numerical example. Consider that $\lambda=0.125$ meter (m) (corresponding to 2.4 GHz wifi) and $T=10^{-3}$ s. Then, $N_0=4$ with $M=500$ lead to the unmeasurable velocity range of $[-0.5,0.5]$ m/s. In general, a large M is advantageous to reduce the unmeasurable velocity range. However, the Doppler frequency may not be constant under a large M. Consider an average radial motion of 2 m/s indoors, $M=500$ leads to a movement of $1(=2\times MT)$ m. The above trade off caused by M needs to be kept in mind when performing coherent Doppler processing.

To proceed, we assume that a notch filter is designed with the null width of N_0 times the Doppler frequency bin $\frac{1}{MT}$. One can easily employ some filter design tools, such as MATLAB [15], to fulfill the task. The details are suppressed here. Applying the filter on $\tilde{\mathbf{h}}_{n,m}$ given in (16), we can obtain

$$\breve{\mathbf{h}}_{n,m} = \mathrm{notchfilter}\left\{\tilde{\mathbf{h}}_{n,m}\right\} \approx \tag{18}$$

$$\begin{cases} \alpha_1\alpha_0^{\dagger}e^{\mathrm{j}2\pi n f_0(\tau_1-\tau_0)}e^{\mathrm{j}2\pi\mu_1 mT}\mathbf{a}_{N_\mathrm{a}}(\theta_1) & \text{if } \mu_1\notin\left[-\frac{N_0\lambda}{2MT},\frac{N_0\lambda}{2MT}\right] \\ 0 & \text{otherwise} \end{cases}.$$

Now, we obtain again a single-tone signal that is over $m=0,1,\cdots,M-1$ with the angular frequency $2\pi\mu_1 T$. Once more, the frequency estimator developed in Section III can be applied for estimating μ_1. Some preparations are needed to make the application more straightforward.

- Extract the following scalar function from $\breve{\mathbf{h}}_{n,m}$,
$$\breve{h}_l[m] = [\breve{\mathbf{h}}_{n,m}]_i,\ l=iN+n,\ i=0,1,\cdots,N_\mathrm{a}-1,$$
$$n=0,1,\cdots,N-1. \tag{19}$$
- Unlike the delay estimation in Algorithm 1, we are now only interested in a limited number of Doppler frequency bins (DFBs), as indoor moving speed is limited. We can safely assume that the maximum speed is ±5 m/s. Then the indexes of DFBs of interest can be given by
$$\mathcal{I}_{\mathrm{DFB}} = \{0,\cdots,x-1,M-x,\cdots,M-1\}, \tag{20}$$

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Algorithm 2 Estimate the Doppler frequency μ_1

Initialize: Set $\delta = 0$, $i_{\text{iter}} = 0$, I_{iter},
Perform Steps 1)-11) of Algorithm 1 with the following changes:

1) Replace $h_{m,i}[n]$ with $\breve{h}_l[m]$, as given in (19);
2) When getting k in Step 3), only consider the DFBs with the indexes in $\mathcal{I}_{\text{DFB}}$ as given in (20);

Return the angular frequency estimate $\widehat{\omega_m^0}$;

TABLE I: Scenario settings

Scenario	Rx Loc. (m)	Tx Loc. (m)	Rx-Orient (rad)
Classroom	$(5.6, 0)$	$(0, 0)$	0.8552
Office	$(2.4, 0)$	$(0, 0)$	0.3665
Corridor	$(1.8, 0)$	$(0, 0)$	0.7854

where $x = \left\lceil \frac{5}{\Delta_v} \right\rceil$, the high end of indexes account for the negative velocity, and Δ_v is given in (17).

Algorithm 2 summarizes the Doppler frequency estimation method. Note that the algorithm estimates the angular frequency of a single-tone signal which is $2\pi\mu_1 T$. Thus, the Doppler frequency estimate is $\widehat{\omega_m^0}/(2\pi T)$, and then the velocity estimate is

$$\hat{v} = \lambda\widehat{\omega_m^0}/(2\pi T), \tag{21}$$

where $\widehat{\omega_m^0}$ is the outcome of Algorithm 2.

IV. Experimental Results

The open wifi CSI data of Widar2.0 [13] is employed to validate the effectiveness of the proposed Doppler estimation method. The experimental platform has a single-antenna wifi transmitter and a three-antenna receiver, hence $N_{\text{a}} = 3$. The CSI capturing interval is $T = 10^{-3}$ s. The carrier frequency is $f_{\text{c}} = 5.825$ GHz, and then the wavelength is $\lambda = 0.0515$ m. The sub-carrier frequency interval is $f_0 = 625$ KHz and the number of sub-carriers per CSI capture is $N = 30$. Experiments based on the above settings were performed for three scenarios in [13], as summarized in Table I. The orient in the table is estimated using target-free CSIs in an initial training stage [13].

Note that the orient is defined as the angle between the receiver array and the x-axis. As illustrated in Fig. 2, both transmitter and receiver lie on the x-axis. Thus, LoS-AoA plus orient angle is equal to 90°. Here, we define AoA as the angle between an incident path onto an array and its normal direction; see Fig. 2. Moreover, when the path is on the left side of the normal direction, the AoA is negative; otherwise, it is positive.

As illustrated in Section III, Algorithm 1 gives one estimate per symbol. Here, we simply take the average of all estimates from the same collection as the final LoS-AoA estimation, as denoted by $\hat{u}_0$. Moreover, we note that $\hat{u}_0$ is the angular frequency, as given by $2\pi d \sin(\theta_0)/\lambda$. Thus, we need a further step to translate $\hat{u}_0$ back to θ_0 estimate, as given by

$$\hat{\theta}_0 = \begin{cases} \text{asin}\{\hat{u}_0/\pi\}, & \text{if } \hat{u}_0 \in [0, \pi] \\ \text{asin}\{(\hat{u}_0 - 2\pi)/\pi\} & \text{if } \hat{u}_0 > \pi \end{cases}. \tag{22}$$

For the case of $\hat{u}_0 > \pi$, a correction is needed; the estimation is in the range of $[0, 2\pi]$, while the true AoA is in the range of $[-\pi, \pi]$. Moreover, we have taken $d = \lambda/2$ in the above calculation.

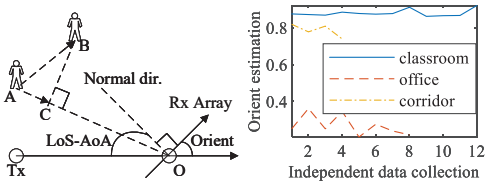


Fig. 2: The geometric relation between wifi Tx and Rx (left). The orient estimation enabled by the proposed Algorithm 1 is plotted on the right.

In the Widar2.0 open data, there are 12, eight and four independent collections with different moving traces for the scenarios of classroom, office and corridor, respectively. Each collection consists of thousands of symbols. Algorithm 1 is performed for each collection per scenario. The results of Step 13) are averaged, which is then translated to the physical AoA per (22). As explained earlier, $\pi/2 - \hat{\theta}_0$ is the orient estimation.

Fig. 2 plots our orient estimation based on trace data (named as T*.mat in Widar2.0 data), in contrast to target-free training data used in Widar2.0 [13]. We see that the results obtained by our methods basically match the orient estimation of Widar2.0, as listed in Table I. The reason for the slightly larger variation in the orient estimates for the office scenario is the so-called zigzag effect. Specifically, the true LoS-AoA (in angular frequency, i.e., $2\pi d \sin(\theta)\lambda$) approaches π in the office scenario; see Fig. 1. Affected by noises and LoS-delay estimation error, the angle estimation can jump over π and is turned into a negative value by the angle correction in (22).

Next, we demonstrate the performance of the proposed Doppler estimation method. Every $M = 200$ symbols are coherently processed to estimate the Doppler frequency, while a gap of 10 symbols is introduced between two Doppler estimates. That is, symbols $0 \sim 199$ are used for the first estimation, symbols $10 \sim 209$ are used for the second, and so on. The same applies to all scenarios.

To evaluate the Doppler estimation quality, we use the ground-truth locations (GTLs) provided by Widar2.0 [13] to derive an approximate radial velocity trace. The GTLs are obtained using video processing techniques, where a video is taken during the data collection and then the locations of the moving body are extracted from the video. Note that we only have two-dimensional body locations; see Figs. 3(a), 3(d) and 3(g) for examples. Assume that the moving speed is constant within each pair of locations and the location sampling time is 1 s[1]. Then, we can estimate the speed between each pair of locations, such as A and B given in Fig. 2, as

$$\vec{AB} \odot \vec{AO}/\|\vec{AO}\|_2, \tag{23}$$

where $\vec{AB}$ denotes a vector pointing from A to B, $\vec{AO}$ points to O from A, $\odot$ denotes the inner product of two vectors, and $\|\|_2$ denotes the ℓ_2-norm. *Due to constant speed assumption, the estimated velocity trace is scaled not only in magnitude but also in the time domain.*

Fig. 3 plots the estimated velocity using the proposed Algorithm 2, in comparison with the velocity trace estimated

[1]The 1 s assumption is to simplify the analysis. Any non-unit time only scales the overall velocity trace by some constant factor.

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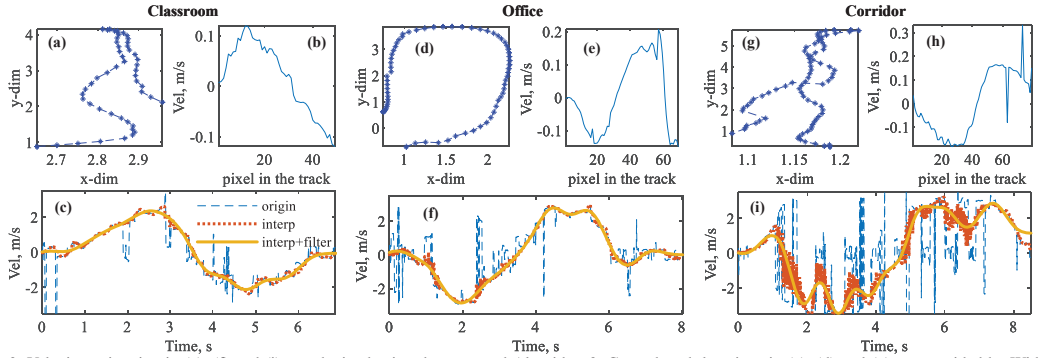


Fig. 3: Velocity estimation in (c), (f) and (i) are obtained using the proposed Algorithm 2. Ground truth locations in (a), (d) and (g) are provided by Widar2.0 [13]. The velocity traces in (b), (e) and (h) are estimated using the ground truth locations; see (23).

from GTLs. Regardless of the scaling effect in time and magnitude of GTL-based velocity traces, the proposed method achieves very similar trends of velocity traces in overall. For all scenarios, the ends of GTL-based velocity traces seem to be incomplete. This may be caused by the large processing interval when extracting the GTLs from the recorded video; similar to signal decimation. The precise recovery of the Doppler traces manifests the feasibility of the overall wifi sensing scheme proposed in this work, i.e., estimating and compensating timing and frequency offsets using the LoS path and then estimating the Doppler frequency through coherent Doppler processing, and the high performance of the proposed Algorithms 1 and 2. Lastly, we highlight that the highly matching Doppler traces also suggest the effective compensation of nuisances caused by clock offsets.

V. Conclusion and Remarks

In this paper, we develop a novel sensing scheme. It starts with estimating the timing and frequency offsets using the LoS path. It then filters out static paths by jointly processing multiple consecutive symbols. It further estimates the Doppler frequency using the filtered signals. We employ the open Widar2.0 data to validate the proposed design. A velocity trace is derived from the GTLs provided in Widar2.0. Our sensing scheme obtains highly matching trends of velocity traces with those estimated from GTLs.

We remark that our design can potentially be extended for estimating multiple moving targets. Specifically, one can see from (3) and (5) that the presence of multiple moving targets shall not affect the proposed estimation methods for LoS parameters. Moreover, one can see from (16) and (18) that multiple moving targets only make the filtered signals become multi-tone signal instead of a single tone. We can invoke estimation-and-subtract strategy to estimate the tones one by one [14]. However, we will need some matching algorithms to associate a Doppler frequency estimate to the correct target.

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